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FINALTERM EXAMINATION
Fall 2009
PHY301- Circuit Theory (Session - 2)

Time: 120 min
Marks: 70

Question No: 1 (Marks: 1) - Please choose one

Charge of 2c and 5c will

- ▶ attract each other
- ▶ **repel each other**
- ▶ no effect
- ▶ cancel each other

Question No: 2 (Marks: 1) - Please choose one

Which battery applies a greater potential difference?

- ▶ **12v**
- ▶ 1.5v
- ▶ 10v
- ▶ 0.5v

Question No: 3 (Marks: 1) - Please choose one

When one resistance in a series string is open

- ▶ The current is maximum in the normal resistances.
- ▶ **The current is zero in all resistances**
- ▶ The voltage is zero across the open resistance
- ▶ The current increases in voltage source

Question No: 4 (Marks: 1) - Please choose one

Addition of impurities in semiconductor material to produce more current is called

- ▶ **dopping**
- ▶ bonding
- ▶ excitation
- ▶ intrinsic

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Waseem.jeetumoni@gmail/yaho/skype/facebook.com

Question No: 5 (Marks: 1) - Please choose one

For a P-N junction under forward bias

- ▶ **more forward current flows**
- ▶ no forward current flows
- ▶ more reverse current flows
- ▶ infinite reverse current flows

Question No: 6 (Marks: 1) - Please choose one

A diode is a

- ▶ **one way conductor**
- ▶ two way conductor
- ▶ three way conductor
- ▶ insulator

Question No: 7 (Marks: 1) - Please choose one

In forward biased Characteristics of junction diode, current value is

- ▶ $i = I_s e^{V/V}$
- ▶ **$I_s = i e^{V/V T}$**
- ▶ $i = I_s e^{V T/V}$
- ▶ $i = I_s e^{V/V T}$

Question No: 8 (Marks: 1) - Please choose one

The primary and secondary winding of transformer are

- ▶ physically touched
- ▶ **physically isolated**
- ▶ touched with conductor
- ▶ largely separated

Question No: 9 (Marks: 1) - Please choose one

A certain transformer has 500 turns in the primary winding and 2500 turns in the secondary winding. The turn ratio is

- ▶ 0.5
- ▶ **25**
- ▶ 25v
- ▶ 2500

Question No: 10 (Marks: 1) - Please choose one

A transformer will be out of phase when

- ▶ output voltage is 0 degree out of phase with input voltage
- ▶ **output voltage is 180 degree out of phase with input voltage**
- ▶ output voltage is 360 degree out of phase with input voltage
- ▶ output voltage is same as input voltage

Question No: 11 (Marks: 1) - Please choose one

If N denotes the total number of nodes, then number of nodal equations in nodal analysis will be

- ▶ **Number of equations=N-1**
- ▶ Number of equations=N-1/2
- ▶ Number of equations=N
- ▶ Number of equations=N-2

Question No: 12 (Marks: 1) - Please choose one

Using superposition theorem, for a circuit containing independent sources, any remaining current source is

- ▶ replaced by capacitor
- ▶ replaced by short circuit
- ▶ **made zero by replacing them by open circuit**
- ▶ replaced by close circuit

Question No: 13 (Marks: 1) - Please choose one

The PIV of a half wave rectifier circuit with a shunt capacitor filter is

- ▶ $2V_m$
- ▶ V_m
- ▶ $V_m/2$
- ▶ $3V_m$

Question No: 14 (Marks: 1) - Please choose one

The primary function of a filter is to

- ▶ Minimize ac input variations
- ▶ Suppress odd harmonics in the rectifier output
- ▶ **Stabilize dc level of the output voltage**
- ▶ Remove ripples from the rectified output

Question No: 15 (Marks: 1) - Please choose one

The base region of a p-n-p transistor is

- ▶ Very thin and heavily doped with holes
- ▶ **Very thin and heavily doped with electrons**
- ▶ Very thin and lightly doped with holes
- ▶ Very thin and lightly doped with electrons

Question No: 16 (Marks: 1) - Please choose one

Voltage multipliers are usually used in low current high voltage applications e.g

- ▶ Stabilizer of refrigerator
- ▶ Microcontroller
- ▶ **Cathode ray tube in television**
- ▶ Remote control

Question No: 17 (Marks: 1) - Please choose one

The current flow across the base-emitter junction of a p-n-p transistor consists of

- ▶ Mainly electrons
- ▶ Equal numbers of holes and electrons
- ▶ **Mainly holes**
- ▶ The leakage current

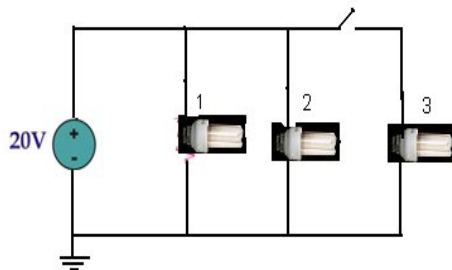
Question No: 18 (Marks: 1) - Please choose one

In an P- type semiconductor material , the majority carriers are

- ▶ **holes**
- ▶ electrons
- ▶ ions
- ▶ protons

Question No: 19 (Marks: 1) - Please choose one

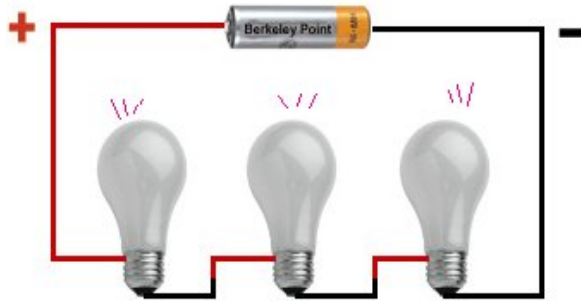
Which light bulb will not glow in given circuit



- ▶ bulb 1 and 2
- ▶ all bulbs will not glow
- ▶ all bulbs will glow
- ▶ **bulb 3**

Question No: 20 (Marks: 1) - Please choose one

Three bulbs are connected in series of a battery, what would happen if any one bulb is opened.



- ▶ half of current will flow
- ▶ same current will flow
- ▶ **no current will flow**
- ▶ $\frac{2}{3}$ current will flow

Question No: 21 (Marks: 3)

A small light bulb with a resistance of 25Ω is connected across the same 220 volt power line. How much is the current 'I'

Solution:

$$R=25\ \Omega$$

$$V=220\ \text{V}$$

From ohm law

$$V=IR, \quad I=V/R$$

So,

$$I=220/25=8.8\ \text{Ampere}$$

Question No: 22 (Marks: 3)

Describe Source Transformation method for simplifying circuit..

Source Transformation:

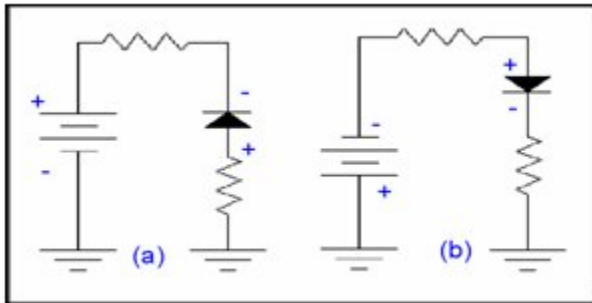
If we have any source embedded within a network, say this source is a current source having a value I & there exists a resistance having a value R , in parallel to it. We can replace it with a voltage source of value $V=IR$ in series with same resistance R .

The reverse is also true that is a voltage source V , in series with a resistance R can be replaced by a current source having a value $I= V/R$ in parallel to the resistance R .

Parameters within circuit, for example an output voltage remain unchanged under these transformations.

Question No: 23 (Marks: 3)

Given below are two figures (a) and (b) having Diode, which diode is forward biased or reversed biased, tell reason.



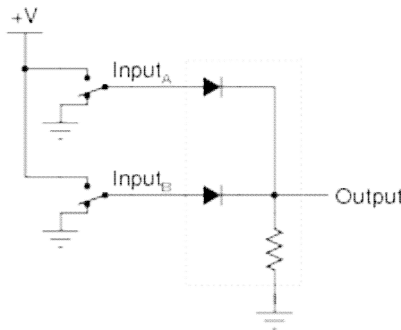
Both A and B are reversed biased

In **fig. a** the negative of the battery is connected to the anode through a resistor and positive terminal of the battery is connected the cathode therefore no current will flow and we can say that diode is not existing and it will act as an open circuit.

In **fig. b** the negative terminal of the battery is connected to the anode through a resistor and positive terminal of the battery is connected the cathode

Question No: 24 (Marks: 3)

Crude logic gates circuits may be constructed out of nothing but diodes and resistors. Take for example this logic gate circuit:



Identify what type of logic function is represented by this gate circuit

This is an OR gate circuit.

Question No: 25 (Marks: 3)

Differentiate between Half wave & Full Wave Rectifier.

Half Wave Rectifier:

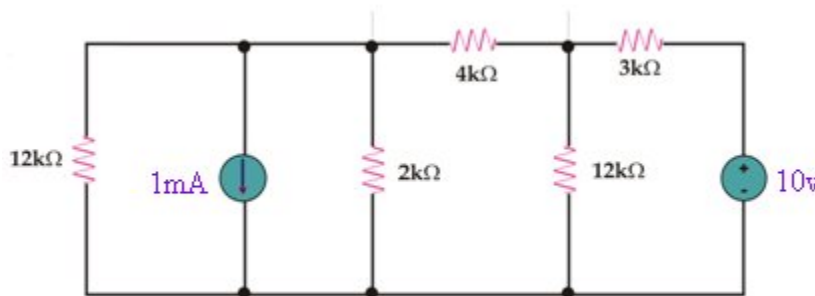
A half wave rectifier is a special case of a clipper. In half wave rectification, either the positive or negative half of the AC wave is passed easily, while the other half is blocked, depending on the polarity of the rectifier. Because only one half of the input waveform reaches the output, it is very inefficient if used for power transfer. Half-wave rectification can be achieved with a single diode in a one phase supply.

Full Wave Rectifier:

A full-wave rectifier converts the whole of the input waveform to one of constant polarity (positive or negative) at its output. Full-wave rectification converts both polarities of the input waveform to DC (direct current), and is more efficient. However, in a circuit with a non-center tapped transformer, four diodes are required instead of the one needed for half-wave rectification.

Question No: 26 (Marks: 5)

Keeping in mind the **Source transformation method**, how we will convert **10v** voltage source into current source and **1mA** into voltage source in the following circuit, Draw diagrams of converted circuit.



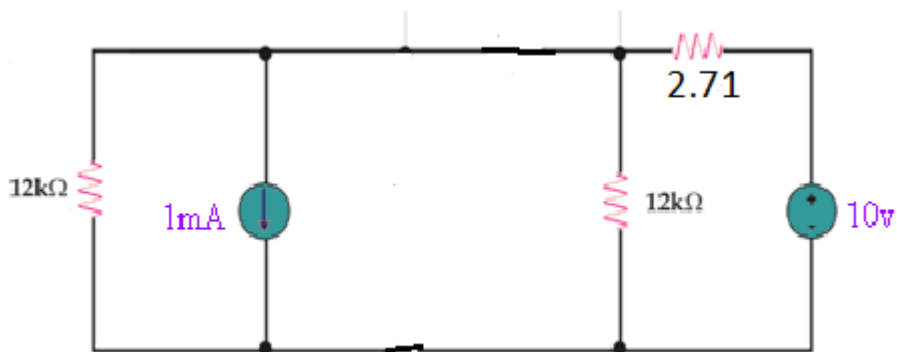
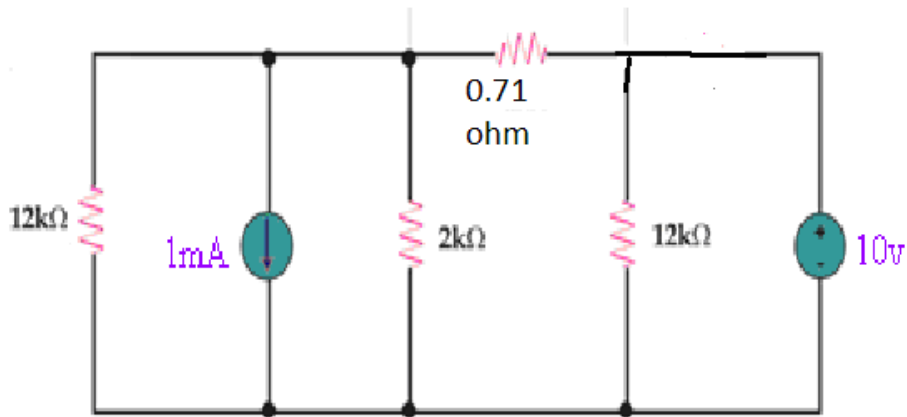
4 is in parallel with 3 so it becomes

$$4 \cdot 3 / 4 + 3 = 1.71 \text{ ohm}$$

0.71 again in series with 2 to give 2.71 ohm

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1mA multiply with 12 k by source transformation to give 12 volts

<http://vuattach.ning.com>

2.71 is in series with 12 to give 12.71 .

10 V is in series with 2.71 so, by source transformation

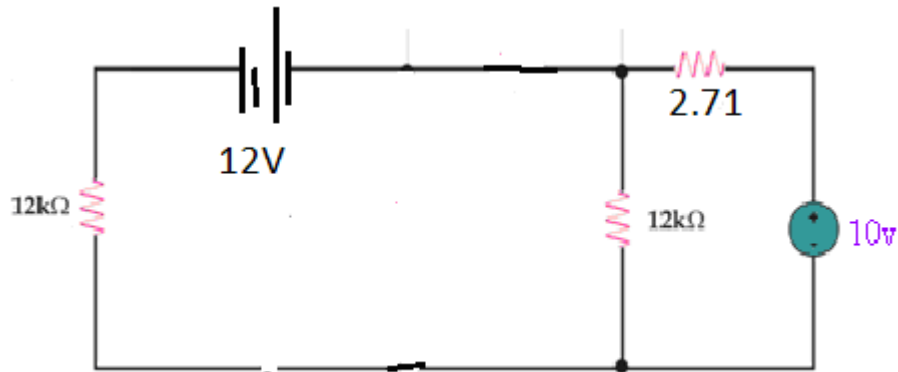
Waseem Jeetumoni

3rd semester

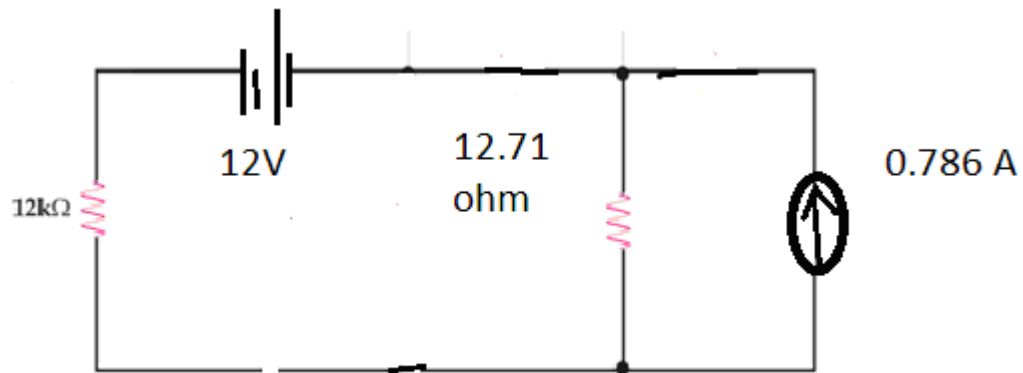
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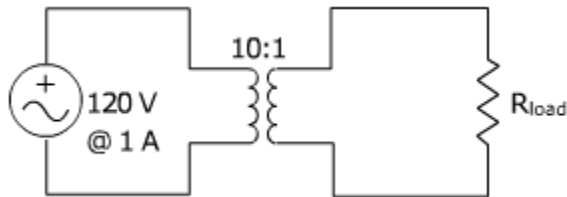
So ,



$$I = 10/12.71 = 0.786A$$

Question No: 27 (Marks: 5)

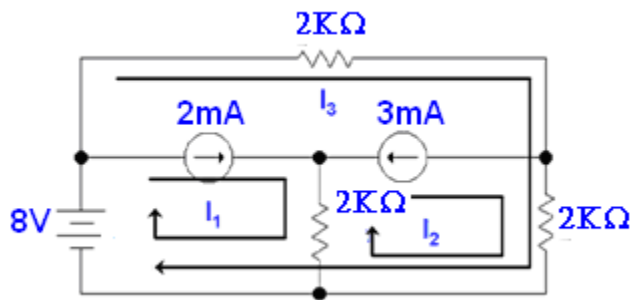
Did the power remain same or not, in either case provide a proof?
<http://vuattach.ning.com>



The power will not remain same. if primary side is equal to 120V then the secondary side will be
 $120/10$
 $=12V$

Question No: 28 (Marks: 5)

Calculate I_1 , I_2 and I_3 values.



Solution:

$$I_1 = 2\text{mA} \dots \text{Ans}$$

$$I_2 = -3\text{mA} \dots \text{Ans}$$

KVL for loop 3

$$2kI_3 + 2k(I_3 + I_2) = 8$$

$$2kI_3 + 2kI_3 + 2kI_2 = 8$$

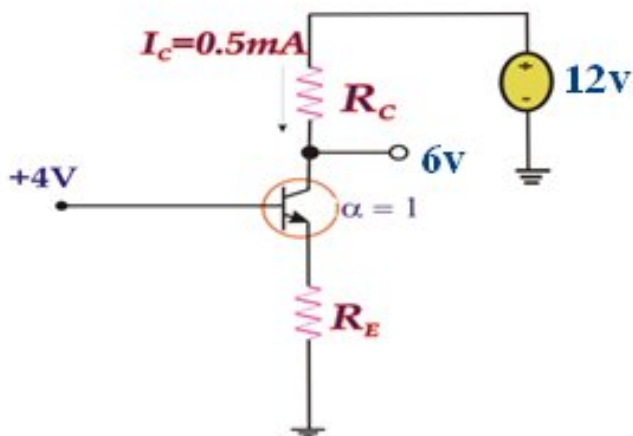
$$4kI_3 - 8 = 8$$

$$4kI_3 = 16$$

$$I_3 = 4\text{mA} \dots \text{Ans}$$

Question No: 29 (Marks: 5)

For the circuit shown, find I_E & R_C



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Question No: 30 (Marks: 5)

A particular diode, for which $n=1$, is found to conduct 3mA with a junction voltage of 0.7V to 0.8V. Find diode current..

Solution:

$$I_d = I_s e^{V_d / nV_t}$$

$$\text{If } I_d = 3 \times 10^{-3} \text{ A}$$

$$\text{when } V_d = 0.7$$

then

$$3 \times 10^{-3} = I_s e^{0.7 / 25.2 \times 10^{-3}}$$

$$= I_s = 2.5905 \times 10^{-15}$$

$$\text{when } V_d = 0.7 \quad I_d = 4.4619 \text{ mA}$$

$$\text{when } V_d = 0.8 \quad I_d = 0.15868 \text{ mA}$$

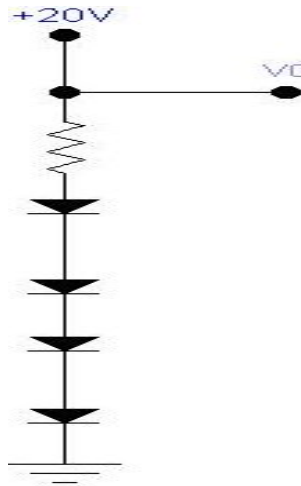
Waseem Jeetumoni

3rd semester

Gujranwala Campus

Question No: 31 (Marks: 10)

Design the circuit in the fig. so that $V_O = 3V$ when $I_L = 0$, and V_O changes by 40mV per 1mA of load current. Find the value of R . (assume four diodes are identical) relative to a diode with 0.7V drop at 1 mA current. Assume $n = 1$



<http://vuattach.ning.com>

SOLUTION: on next:

$V_O = 3V$, when $I_L = 0$, therefore each diode should exhibit a drop of 0.75V. If $I_L = 1mA$, then V_O changes by 40mV and a change due to each diode is 10mV.

Hence

$$r_d = 10mV/1mA = 10 \text{ Ohms}$$

but

$$r_d = nV_T/I_D$$

$$10 = 1 \times 25m/I_D$$

$$I_D = 2.5mA$$

Hence

$$15 - 3 - I_D R_D = 0$$

$$R = (15 - 3)/I_D = (15 - 3)/2.5 = 4.8k \text{ Ohms}$$

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PHY301-MCQs from Final
Term Past Papers Solved
with References

By Masoom Fairy

FINAL TERM EXAMINATION

Spring 2010

PHY301- Circuit Theory

Time: 90 min

Marks: 60

Question No: 1 (Marks: 1) - Please choose one

If we connect 3 capacitors in parallel, the combined effect of all these capacitors will be

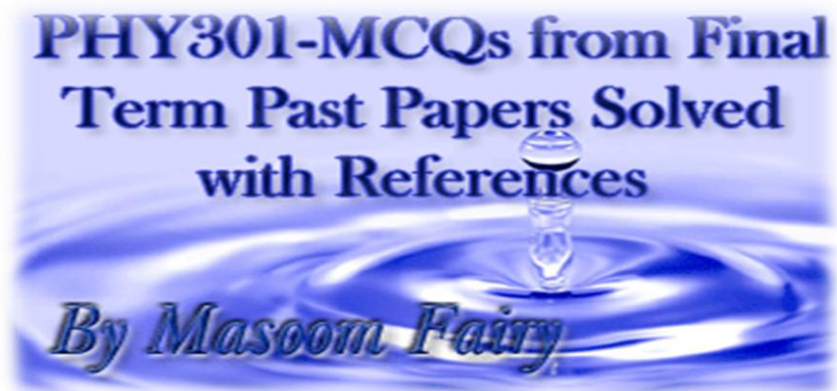
- ▶ **Equal to the sum of individual capacitances (PAGE 22)**
- ▶ sum of reciprocals of individual capacitors
- ▶ zero
- ▶ product of all

Question No: 2 (Marks: 1) - Please choose one

In an N- type semiconductor material , the majority carrier are

- ▶ holes
- ▶ **electrons (PAGE 122)**
- ▶ ions
- ▶ protons

Question No: 3 (Marks: 1) - Please choose one



Semiconductor element in pure form is called

- ▶ extrinsic
- ▶ P type
- ▶ **intrinsic (PAGE 121)**
- ▶ N type

Question No: 4 (Marks: 1) - Please choose one

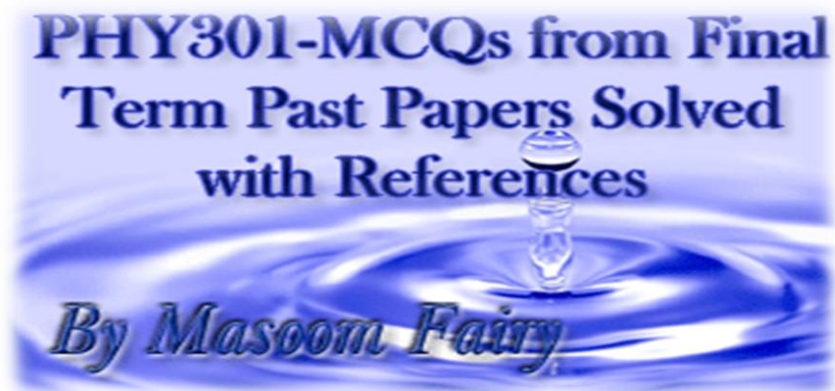
An element with 6 protons and 6 neutrons has atomic no. value

- ▶ **6 (PAGE 5)**
- ▶ 12
- ▶ 18
- ▶ 8

Question No: 5 (Marks: 1) <http://vustudents.ning.com>- Please choose one

Isotopes of an element have

- ▶ **Different mass number but same atomic number**
- ▶ Different atomic number but same mass number
- ▶ Same atomic and mass number
- ▶ Same no. of neutron



[http://wiki.answers.com/Q/Do isotopes have the same atomic number but different mass numbers](http://wiki.answers.com/Q/Do_isotopes_have_the_same_atomic_number_but_different_mass_numbers)

Question No: 6 (Marks: 1) - Please choose one

Which of the following is true about the electrical circuit in your flashlight?

▶ **The battery supplies the charge (electrons) which move through the wires.**

- ▶ The battery supplies the charge (protons) which move through the wires
- ▶ The battery supplies energy which raises charge from low to high voltage
- ▶ Battery gets energy from circuit.

Question No: 7 (Marks: 1) - Please choose one

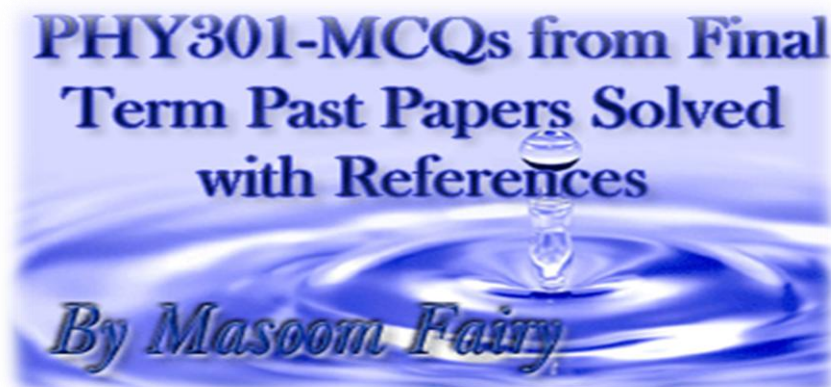
If we connect 3 capacitors in series, the combined effect of all these capacitors will be

- ▶ equal to the sum of individual capacitors
- ▶ **the sum of reciprocals of individual capacitors (PAGE 22)**
- ▶ zero
- ▶ product of all

Question No: 8 (Marks: 1) - Please choose one

which of following relation is true for turn ratio and voltage ratio of transformer

- ▶ $N_2/N_1 = V_1/V_2$



► $N_2/N_1 = V_1/V_2$

► $N_1/N_2 = V_1V_2/V_2$

► $N_2/N_1 = V_2/V_1$ (PAGE 144)

Question No: 9 (Marks: 1) - Please choose one

Leakage current of semiconductor diode is caused by

► chemical energy

► heat energy

► barrier voltage

► **doping impurity**

Question No: 10 (Marks: 1) - Please choose one

Rectification is the process of

► converting DC into AC

► increasing output waveform

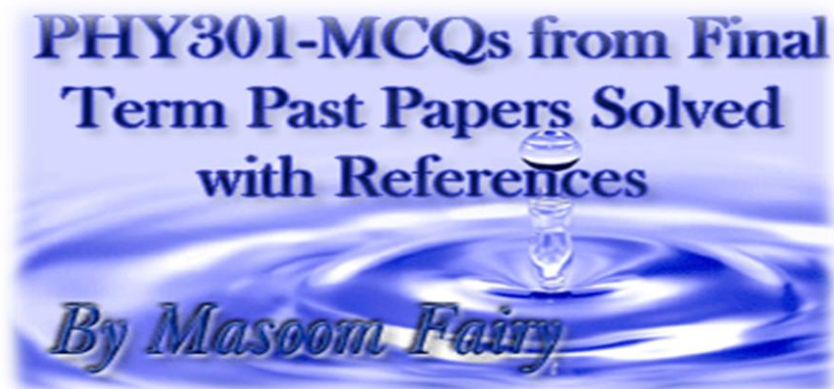
► **converting AC into DC (PAGE 126)**

► converting diode into capacitor

Question No: 11 (Marks: 1) - Please choose one

which rectifier is used to get either positive or negative part of input at output

► **half wave rectifier (PAGE 147)**



- ▶ full wave rectifier
- ▶ bridge rectifier
- ▶ negative half wave rectifier

Question No: 12 (Marks: 1) - Please choose one

The reference/common point among all the nodes without insertion of any component between them is called

- ▶ node
- ▶ **ground (PAGE 42)**
- ▶ loop
- ▶ mesh

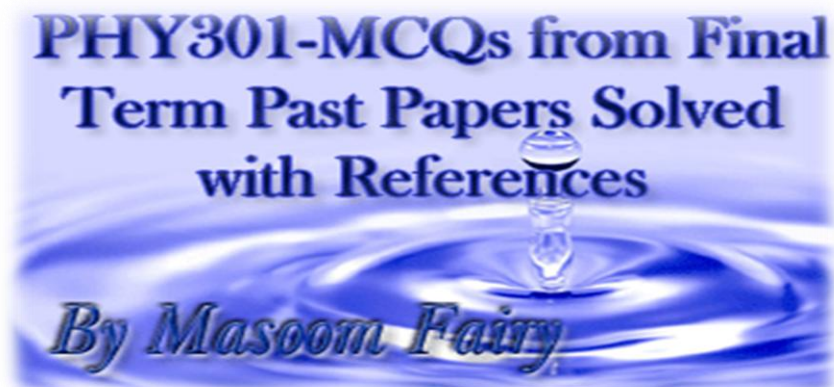
Question No: 13 (Marks: 1) - Please choose one

In a Norton's theorem , norton's resistance **Inor** is calculated

- ▶ at open terminals of the open circuit
- ▶ **by short circuiting the open terminal of the circuit (PAGE 113)**
- ▶ across the load
- ▶ any where

Question No: 14 (Marks: 1) - Please choose one

In a thevenin's theorem , the condition to find V_{th} is



- ▶ to add more load resistance
- ▶ to remove load resistance (PAGE 100)
- ▶ to convert load into short circuit
- ▶ open circuit the current source

Question No: 15 (Marks: 1) - Please choose one

Using superposition theorem, for a circuit containing independent sources, any remaining voltage source is

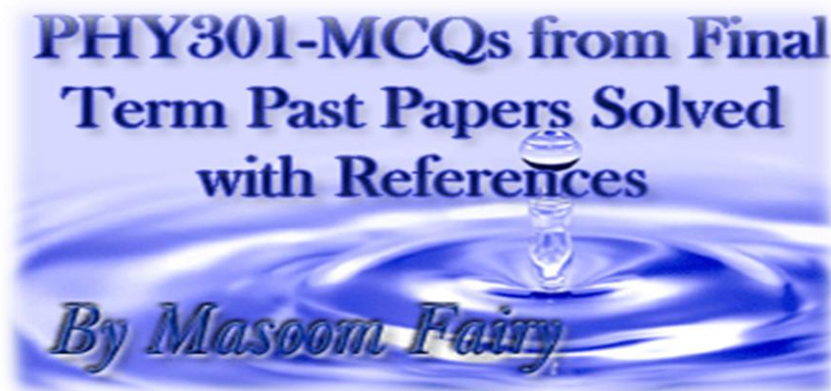
- ▶ remain same
- ▶ made zero by replacing them by open circuit
- ▶ made zero by replacing them by capacitor
- ▶ made zero by replacing them by short circuit (PAGE 92)

Question No: 16 (Marks: 1) - Please choose one

In Norton, s theorem , while calculating R_n

- ▶ Short circuit the current source and open circuit the voltage source
- ▶ open circuit the current source and short circuit the voltage source (PAGE 114 Example number 1 Step3)
- ▶ insert the load
- ▶ just open circuit current source

Question No: 17 (Marks: 1) - Please choose one



The primary function of a filter is to

- ▶ Minimize ac input variations
- ▶ Suppress odd harmonics in the rectifier output
- ▶ Stabilize dc level of the output voltage
- ▶ **Remove ripples from the rectified output (PAGE 159)**

Question No: 18 (Marks: 1) - Please choose one

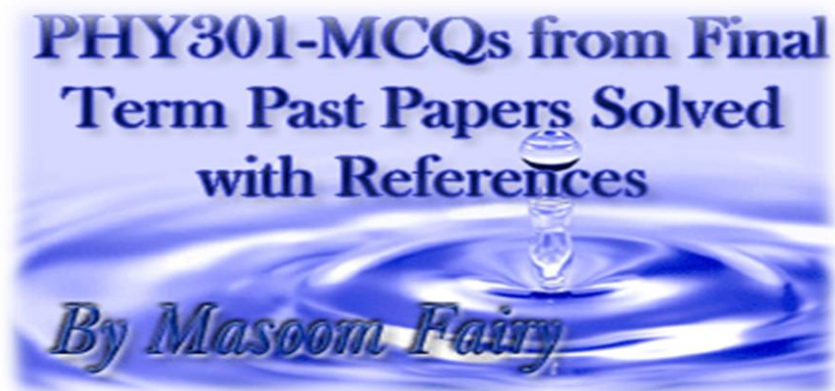
The PIV of a half wave rectifier circuit with a shunt capacitor filter is

- ▶ $2V_m$
- ▶ **V_m (PAGE 161)**
- ▶ $V_m/2$
- ▶ $3V_m$

Question No: 19 (Marks: 1) - Please choose one

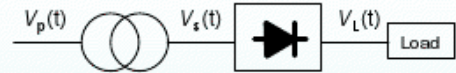
The ripple factor of a bridge rectifier

- ▶ 1.21
- ▶ **0.406 (Not Conform)**
- ▶ 0.812
- ▶ 1.11



Comparison among topologies 1-Ph.

- Secondary voltage is sinusoidal: $v_s(t) = V_s \sin(2\pi f_{\text{ mains }} t)$
- Resistive Load
- Ideal devices (no device losses)



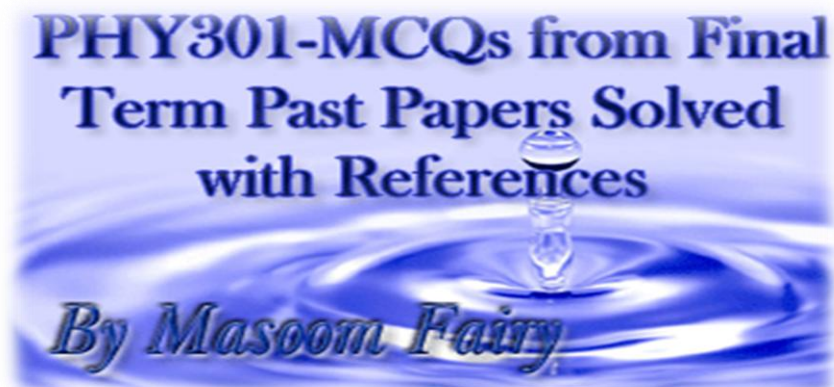
Parameter	Half-Wave	Full - Wave (Center-tapped)	Full - Wave (Bridge)
Rectified Voltage - V_{DC}	$V_s / \pi = 0.318 \cdot V_s$	$2 \cdot V_s / \pi = 0.636 \cdot V_s$	$2 \cdot V_s / \pi = 0.636 \cdot V_s$
rms Output Voltage - V_L	$V_s / 2 = 0.318 \cdot V_s$	$V_s / \sqrt{2} = 0.707 \cdot V_s$	$V_s / \sqrt{2} = 0.707 \cdot V_s$
Form Factor - FF	1.57	1.11	1.11
Rectification Ratio - η	0.405	0.81	0.81
Ripple Factor - RF	1.21	0.482	0.482
Transformer Utilization Factor - TUF	0.286	0.572	0.81
Diode Peak Inverse Voltage (PIV) - V_{PRM}	$V_s = \pi \cdot V_{DC}$	$2 \cdot V_s = \pi \cdot V_{DC}$	$V_s = \pi/2 \cdot V_{DC}$
Peak Direct Voltage (PDV - thyristors only) - V_{DRM}	$V_s = \pi \cdot V_{DC}$	$2 \cdot V_s = \pi \cdot V_{DC}$	$V_s = \pi/2 \cdot V_{DC}$
Diode Peak Forward Current - I_{PRM}	$\pi \cdot I_{DC}$	$\pi/2 \cdot I_{DC}$	$\pi/2 \cdot I_{DC}$
Diode Average Current - $I_{F(AV)}$	I_{DC}	$0.5 \cdot I_{DC}$	$0.5 \cdot I_{DC}$
Diode Rms Current - $I_{F(RMS)}$	$\pi/2 \cdot I_{DC}$	$\pi/4 \cdot I_{DC}$	$\pi/4 \cdot I_{DC}$
Fundamental Ripple Frequency - f_R	$f_{\text{ mains }}$	$2 \cdot f_{\text{ mains }}$	$2 \cdot f_{\text{ mains }}$

Phy301-Solved papers with reference by Masoom fairy

Question No: 20 (Marks: 1) - Please choose one

Zener breakdown occurs

- Mostly in germanium junctions
- Due to rupture of covalent bond
- In lightly doped-junctions



- ▶ Due to thermally –generated minority carriers

In Zener Breakdown, the breakdown is initiated by the rupture of the bonds because of the existence of a strong electric field.

<http://www.allinterview.com/showanswers/10620.html>

Question No: 21 (Marks: 1) - Please choose one

The clipping level is primarily determined by

- ▶ Shape of the input waveform
- ▶ Battery voltage
- ▶ Value of the resistor
- ▶ **Knee voltage of the diode (PAGE 160)**

(2) Determine the applied voltage (Transition Voltage) that causes change in the diode bias.

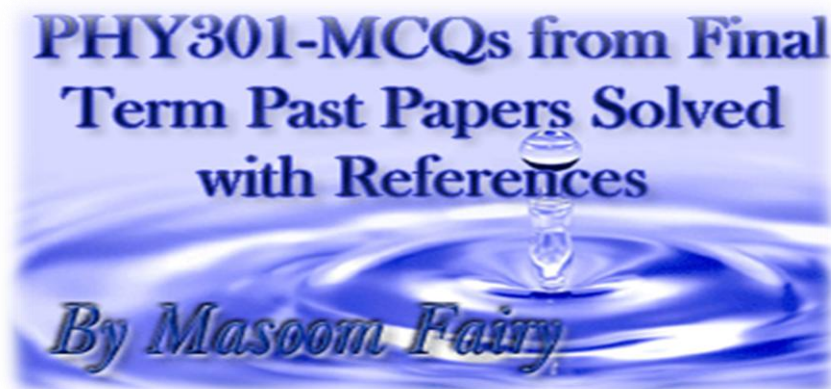
Question No: 22 (Marks: 1) - Please choose one

Without a DC source a clipper acts like a

- ▶ **Rectifier**
- ▶ Clamper
- ▶ Demodulator
- ▶ Chopper

http://ugcelectronics.blogspot.com/2012/05/electronic-science-solved-question_07.html

Question No: 23 (Marks: 1) - Please choose one



Avalanche breakdown is primarily dependent on the phenomenon of

► **Collision**

► Doping

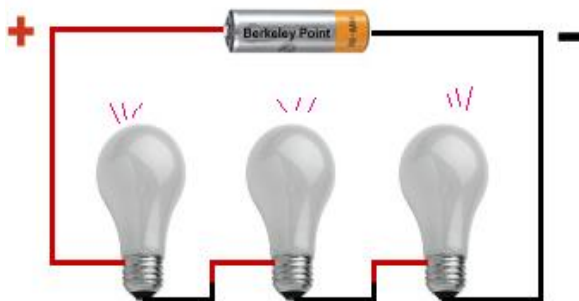
► Ionization

► Recombination

http://wiki.answers.com/Q/What_is_Avalanche_breakdown

Question No: 24 (Marks: 1) - Please choose one

Three bulbs are connected in series of a battery, what would happen if any one bulb is opened.

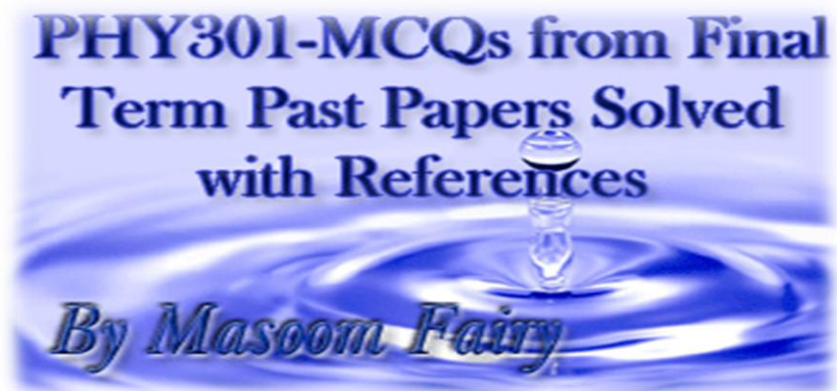


► half of current will flow

► same current will flow

► **no current will flow**

► 2/3 current will flow



In a series circuit there is only one path for current to flow. The current will flow through each good lamp. If any lamp opens (blows), then the circuit is broken and current flow stops.

[http://wiki.answers.com/Q/If one of the three lamps blows out when connected in series what happens to the current in the other two](http://wiki.answers.com/Q/If_one_of_the_three_lamps_blow_out_when_connected_in_series_what_happens_to_the_current_in_the_other_two)

Question No: 25 (Marks: 1) - Please choose one

The base region of a p-n-p transistor is

- ▶ **Very thin and heavily doped with holes**
- ▶ Very thin and heavily doped with electrons
- ▶ Very thin and lightly doped with holes
- ▶ Very thin and lightly doped with electrons

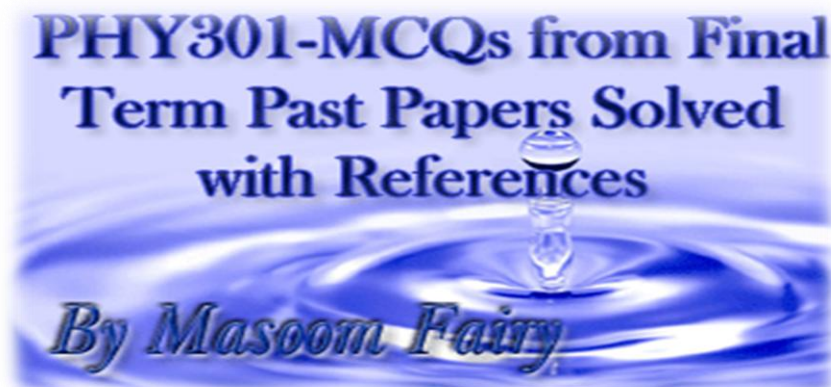
P-N-P Types

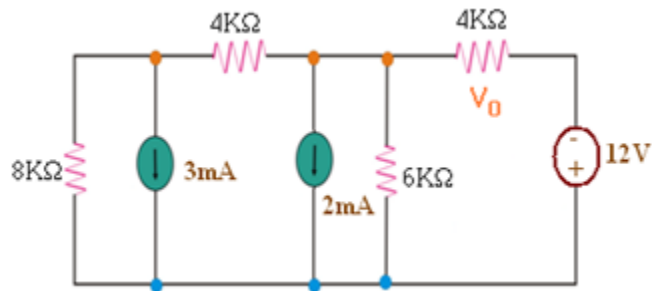
The operation of a p-n-p transistor is essentially identical to that of an n-p-n transistor except that the polarities of the bias voltages are reversed and the main current carriers are holes instead of electrons.

<http://www.eteonline.com/npnnpnp.html>

Question No: 26 (Marks: 1) - Please choose one

Converting 12v voltage source into current source, value of converted current source will be





► 3mA

► 2mA

► 48mA

► 1mA

$$I = V/R \Rightarrow = 12/4 = 3\text{mA}$$

FINAL TERM EXAMINATION

Fall 2009

PHY301- Circuit Theory (Session - 2)

Time: 120 min

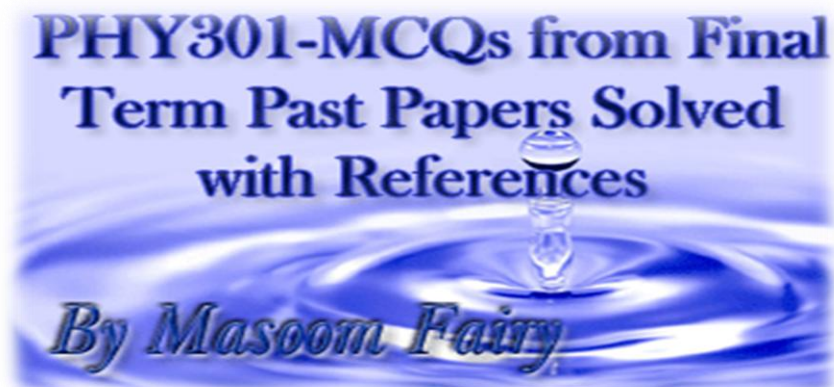
Marks: 70

Question No: 1 (Marks: 1) - Please choose one

Charge of 2c and 5c will

► attract each other

► repel each other (PAGE 9)



- ▶ no effect
- ▶ cancel each other

Question No: 2 (Marks: 1) - Please choose one

Which battery applies a greater potential difference?

▶ **12v (PAGE 8)**

- ▶ 1.5v
- ▶ 10v
- ▶ 0.5v

Question No: 3 (Marks: 1) - Please choose one

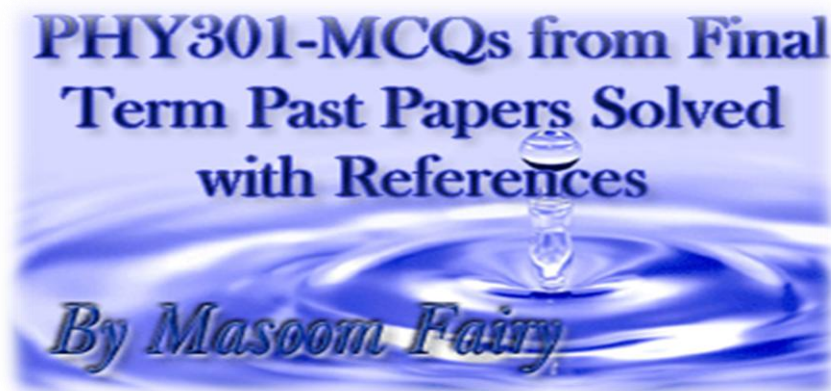
When one resistance in a series string is open

- ▶ The current is maximum in the normal resistances.
- ▶ **The current is zero in all resistances (Lecture 5 Topic: Voltage divider)**
- ▶ The voltage is zero across the open resistance
- ▶ The current increases in voltage source

Reference: Series circuit is Voltage divider so the voltage will exist into the circuit and Current will become zero.

Question No: 4 (Marks: 1) - Please choose one

Addition of impurities in semiconductor material to produce more current is called



► **dopping (During Video Lecture 29)**

- bonding
- excitation
- intrinsic

Question No: 5 (Marks: 1) - Please choose one

For a P-N junction under farward bias

► **more farward current flows (PAGE 176)**

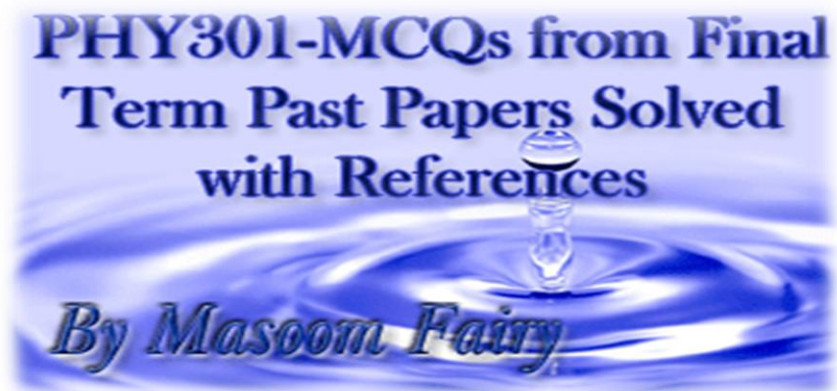
- no farward current flows
- more reverse current flows
- infinite reverse current flows

Question No: 6 (Marks: 1) - Please choose one

A diode is a

► **one way conductor (Lecture 30)**

- two way conductor
- three way conductor
- insulator



Question No: 7 (Marks: 1) - Please choose one

In forward biased Characteristics of junction diode, current value is

- ▶ $i = I_s e^{V/V}$
- ▶ $I_s = i e^{V/V T}$
- ▶ $i = I_s e^{V T/V}$
- ▶ $i = I_s e^{V/V T}$ (PAGE 129)

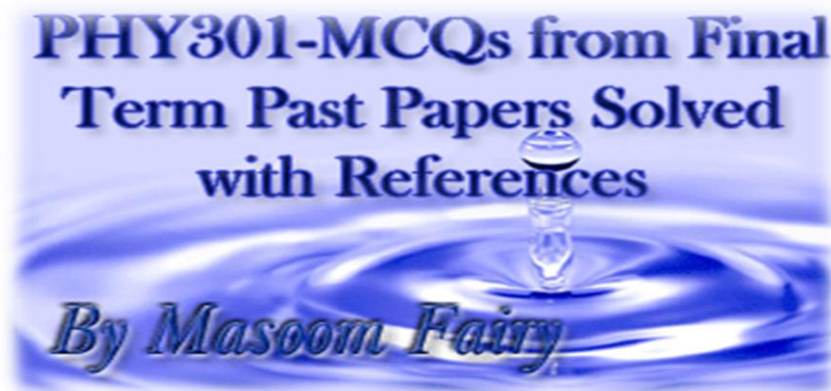
Question No: 8 (Marks: 1) - Please choose one

The primary and secondary winding of transformer are

- ▶ physically touched
- ▶ physically isolated (PAGE 143)
- ▶ touched with conductor
- ▶ largely separated

Question No: 9 (Marks: 1) - Please choose one

A certain transformer has 500 turns in the primary winding and 2500 turns in the secondary winding. The turn ratio is



▶ 0.5

▶ 25

▶ 25v

▶ 2500

$$N_2/N_1 = 2500/500 = 5$$

Question No: 10 (Marks: 1) - Please choose one

A transformer will be out of phase when

▶ output voltage is 0 degree out of phase with input voltage

▶ output voltage is 180 degree out of phase with input voltage (PAGE 146)

▶ output voltage is 360 degree out of phase with input voltage

▶ output voltage is same as input voltage

Question No: 11 (Marks: 1) - Please choose one

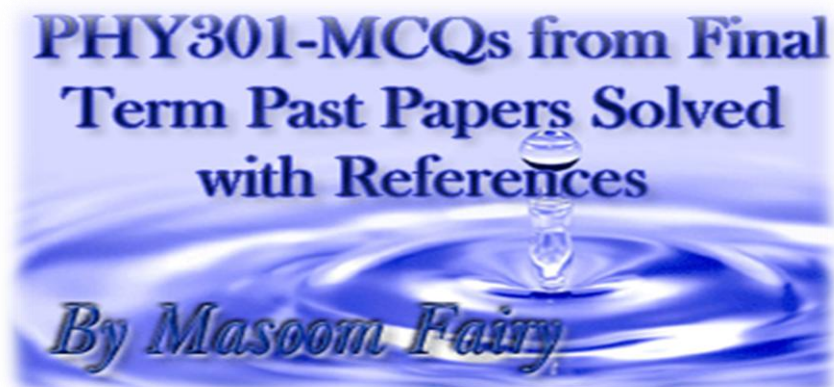
If N denotes the total number of nodes, then number of nodal equations in nodal analysis will be

▶ Number of equations=N-1 (PAGE 42)

▶ Number of equations=N-1/2

▶ Number of equations=N

▶ Number of equations=N-2



Question No: 12 (Marks: 1) - Please choose one

Using superposition theorem, for a circuit containing independent sources, any remaining current source is

- ▶ replaced by capacitor
- ▶ replaced by short circuit
- ▶ **made zero by replacing them by open circuit (PAGE 92)**
- ▶ replaced by close circuit

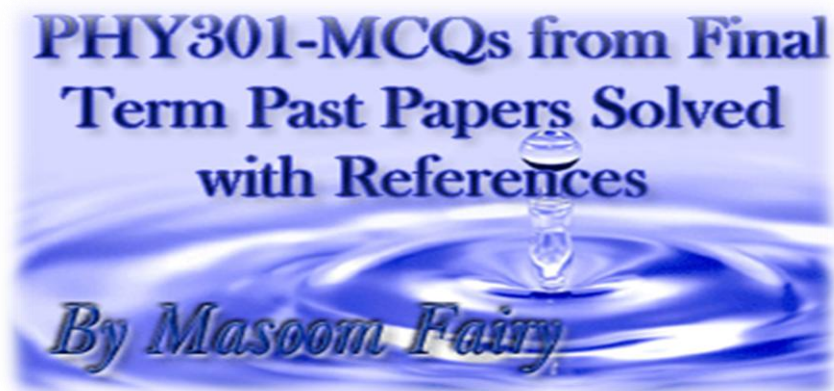
Question No: 13 (Marks: 1) - Please choose one

The PIV of a half wave rectifier circuit with a shunt capacitor filter is

- ▶ $2V_m$
- ▶ **V_m (PAGE 161 Repeated)**
- ▶ $V_m/2$
- ▶ $3V_m$

Question No: 14 (Marks: 1) - Please choose one

The primary function of a filter is to



- ▶ Minimize ac input variations
- ▶ Suppress odd harmonics in the rectifier output
- ▶ **Stabilize dc level of the output voltage**
- ▶ **Remove ripples from the rectified output (PAGE 159 Repeated)**

Question No: 15 (Marks: 1) - Please choose one

The base region of a p-n-p transistor is

- ▶ **Very thin and heavily doped with holes (REPEATED)**
- ▶ Very thin and heavily doped with electrons
- ▶ Very thin and lightly doped with holes
- ▶ Very thin and lightly doped with electrons

P-N-P Types

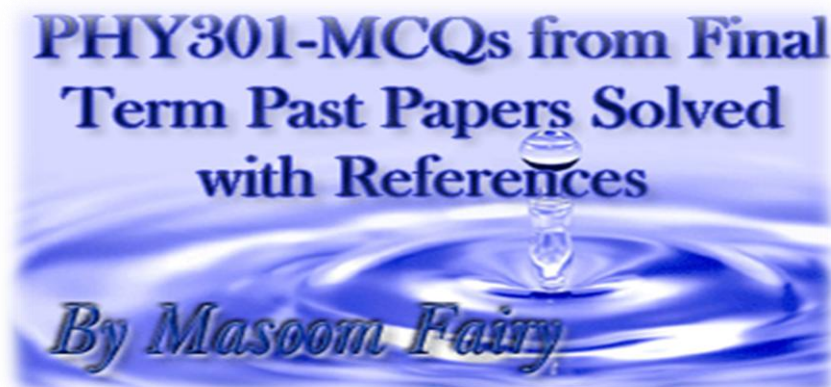
The operation of a p-n-p transistor is essentially identical to that of an n-p-n transistor except that the polarities of the bias voltages are reversed and the main current carriers are holes instead of electrons.

<http://www.eteonline.com/npnnpnp.html>

Question No: 16 (Marks: 1) - Please choose one

Voltage multipliers are usually used in low current high voltage applications e.g

- ▶ Stabilizer of refrigerator



- ▶ Microcontroller
- ▶ Cathode ray tube in television (PAGE 168)
- ▶ Remote control

Question No: 17 (Marks: 1) - Please choose one

The current flow across the base-emitter junction of a p-n-p transistor consists of

- ▶ Mainly electrons
- ▶ Equal numbers of holes and electrons
- ▶ Mainly holes
- ▶ The leakage current

P-N-P Types

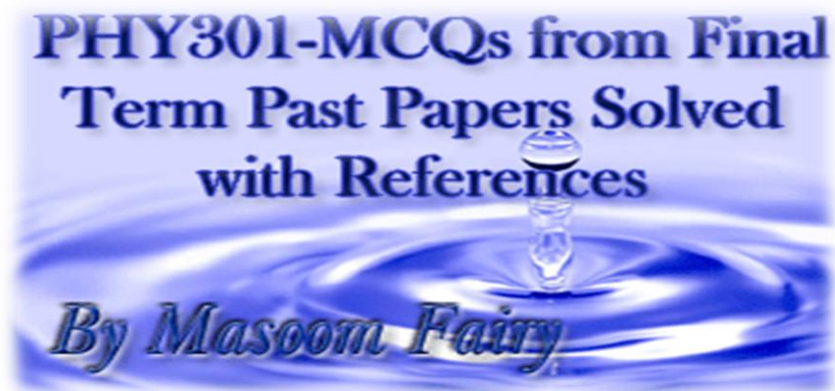
The operation of a p-n-p transistor is essentially identical to that of an n-p-n transistor except that the polarities of the bias voltages are reversed and the main current carriers are holes instead of electrons.

<http://www.eteonline.com/npnpnp.html>

Question No: 18 (Marks: 1) - Please choose one

In an P- type semiconductor material , the majority carriers are

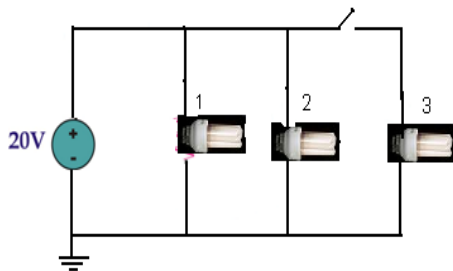
- ▶ holes (PAGE 122)



- ▶ electrons
- ▶ ions
- ▶ protons

Question No: 19 (Marks: 1) - Please choose one

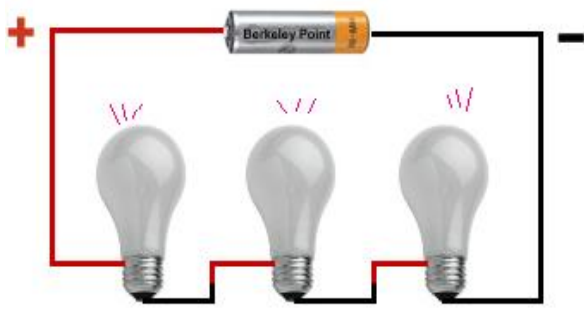
Which light bulb will not glow in given circuit



- ▶ bulb 1 and 2
- ▶ all bulbs will not glow
- ▶ all bulbs will glow
- ▶ **bulb 3**

Question No: 20 (Marks: 1) - Please choose one

Three bulbs are connected in series of a battery, what would happen if any one bulb is opened.



s from Final
ers Solved
ences

By Masoom Fairy

- ▶ half of current will flow
- ▶ same current will flow
- ▶ **no current will flow (REPEATED)**
- ▶ $\frac{2}{3}$ current will flow

In a series circuit there is only one path for current to flow. The current will flow through each good lamp. If any lamp opens (blows), then the circuit is broken and current flow stops.

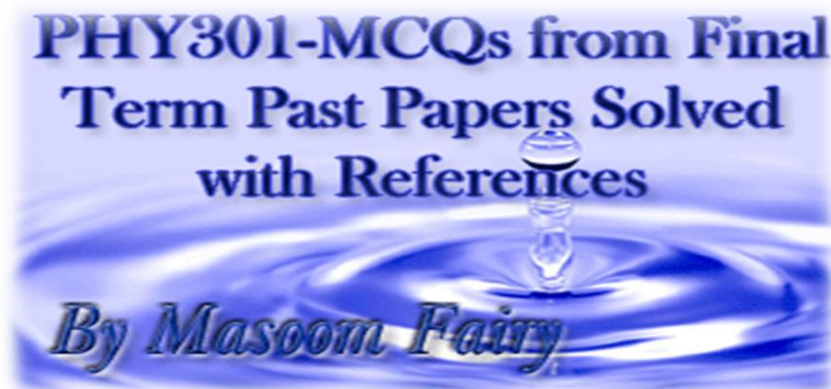
[http://wiki.answers.com/Q/If one of the three lamps blows out when connected in series what happens to the current in the other two](http://wiki.answers.com/Q/If_one_of_the_three_lamps_blow_out_when_connected_in_series_what_happens_to_the_current_in_the_other_two)

Question No: 21 (Marks: 1) - Please choose one

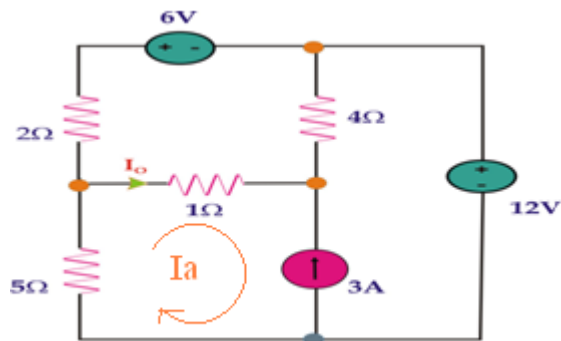
In normal operation an n-p-n transistor connected in common-base configuration has

- ▶ The emitter at a lower potential than the base
- ▶ The collector at a lower potential than the base
- ▶ **The base at a lower potential than the emitter (Not Conform)**
- ▶ The collector at a lower potential than the emitter

Question No: 22 (Marks: 1) - Please choose one



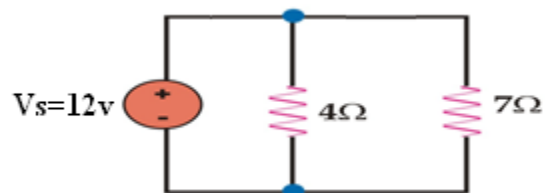
The value of I_a for given circuit is



- $I_a = I_o$
- $I_a = 3A$
- $I_a = -3A$ (3A is in opposite direction)
- $I_a = 6V$

Question No: 23 (Marks: 1) - Please choose one

Across which resistance more voltage is dropped



- 4Ω

► 7Ω

► same across both (PAGE 32)

► no voltage drop

Any Series Circuit is Voltage divider and Parallel Circuit is Current divider.

Question No: 24 (Marks: 1) - Please choose one

When a diode is reversed biased

► It will act as short circuit

► It will act as an open circuit (Video Lecture 30)

► It will act as close circuit

► maximum current flows

FINALTERM EXAMINATION

Spring 2005

PHY301- Circuit Theory

For each question given below, encircle the option that in your opinion represents the best/ correct answer.

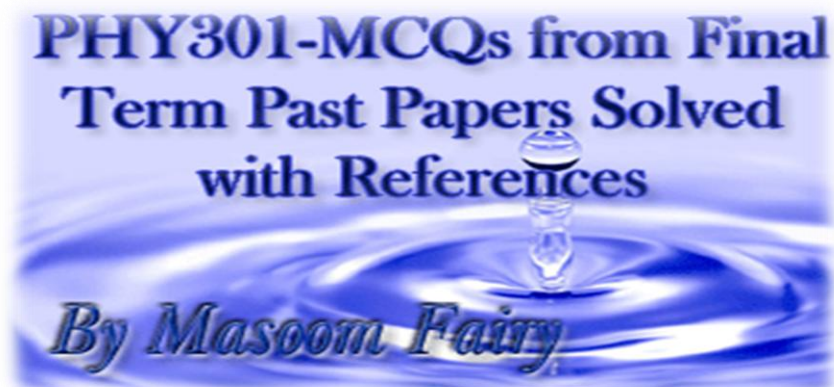
1)The resistance in a short circuit is

a) Large

b) small

c) Zero (PAGE 11)

d) None of the above



2) If one of the diode in a full -wave bridge rectifier is open, the out put is

- a) 0V
- b) One -fourth the amplitude of the input voltage
- c) a half -wave rectified voltage
- d) a 120 HZ voltage

3) The charge in motion is called

- a) Voltage
- b) current (PAGE 8)
- c) intensity
- d) Coulomb

4) The emitter current is always

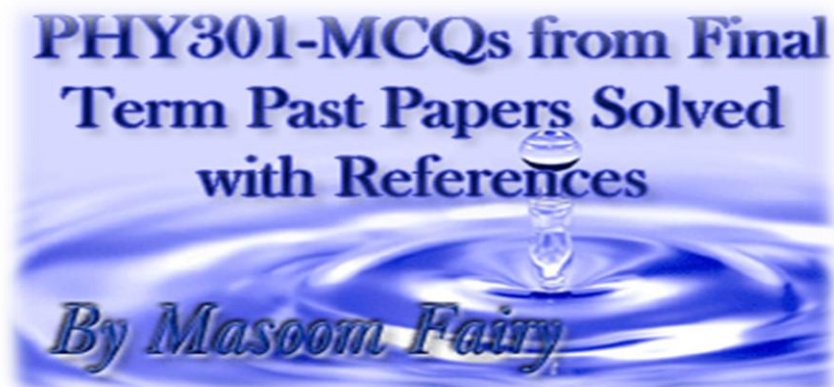
- a) Greater than base current
- b) Less than collector current
- c) Greater than collector current (PAGE 180)
- d) Small than base current

5) In PNP transistor, the P-regions are

- a) Base and emitter
- b) Base and collector
- c) emitter and collector (PAGE 176)
- d) All of the above

FINAL TERM EXAMINATION

Spring 2011



PHY301- Circuit Theory

Leakage current of semiconductor diode is caused by

[marks1]

- ▶ chemical energy
- ▶ heat energy
- ▶ barrier voltage
- ▶ **doping impurity**

Using superposition theorem, for a circuit containing independent sources, any remaining voltage source is

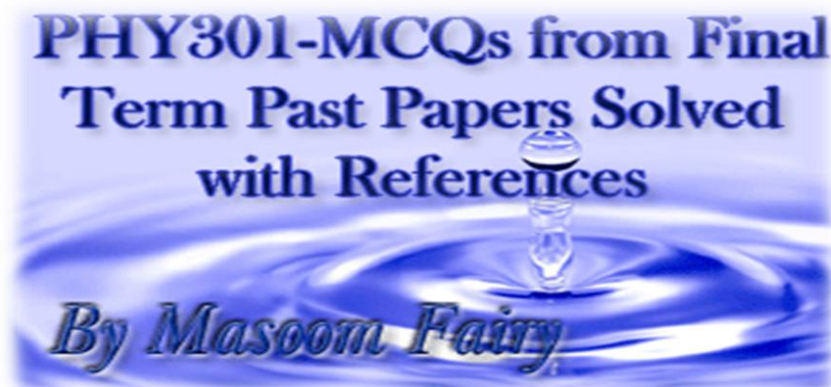
[marks1]

- ▶ remain same
- ▶ made zero by replacing them by open circuit
- ▶ made zero by replacing them by capacitor
- ▶ **made zero by replacing them by short circuit (PAGE 92) Repeated**

The primary function of a filter is to

[marks1]

- ▶ Minimize ac input variations
- ▶ Suppress odd harmonics in the rectifier output
- ▶ Stabilize dc level of the output voltage
- ▶ **Remove ripples from the rectified output (PAGE 159) Repeated**



The ripple factor of a bridge rectifier

[marks1]

- ▶ 1.21
- ▶ 0.406 Repeated
- ▶ 0.812
- ▶ 1.11

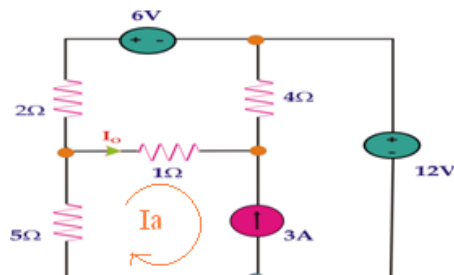
Avalanche breakdown is primarily dependent on the phenomenon of [marks1]

- ▶ Collision (Repeated)
- ▶ Doping
- ▶ Ionization
- ▶ Recombination

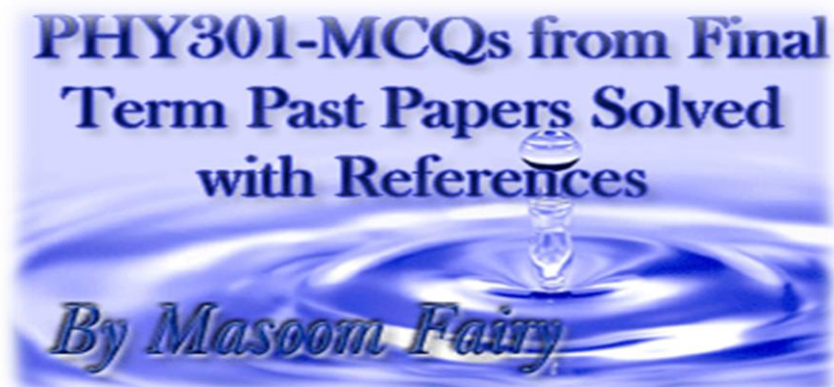
http://wiki.answers.com/Q/What_is_Avalanche_breakdown

The value of I_a for given circuit is

[marks1]



- ▶ $I_a = I_o$
- ▶ $I_a = 3A$



► $I_a = -3A$ (3A is in opposite direction) Repeated

► $I_a = 6V$

When a diode is reversed biased

[marks1]

► It will act as short circuit

► It will act as an open circuit (Video Lecture 30)

► It will act as close circuit

► maximum current flows

FINAL TERM EXAMINATION

Spring 2004

PHY301- Circuit Theory

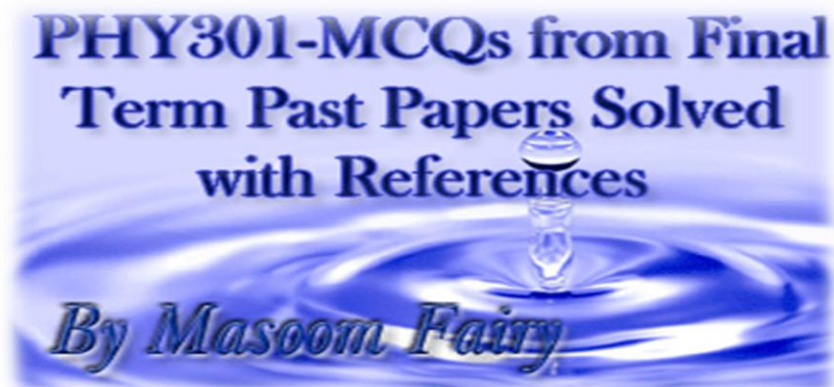
For each question given below, encircle the option that in your opinion represents the best/ correct answer.

I) Choose the correct term, an electronic device that is capable of storing a charge for a long time.

- a) Transistor
- b) Semi-conductor
- c) Capacitor
- d) Diode

II) The unit of electrical potential difference is

- a) watt
- b) Volt (PAGE 8)
- c) ohm
- d) ampere



III) Resistances R1 and R2 are in series with 90 V applied. If V1 is 30 V then V2 must be

- a) 30 V
- b) 90 V
- c) 45 V
- d) 60 V (PAGE 34)

Reference:

$$V_t = 90\text{v} \Rightarrow V_1 = 30\text{v}$$

$$V_t - V_1 = V_2 \Rightarrow 90\text{v} - 30\text{v} = 60\text{v}$$

IV) In an N –type semiconductor materials, the majority carrier are

- a) Holes
- b) Electrons (PAGE 122)
- c) Ions
- d) Protons

V) When a current of 20 milliamps flows through a 50 ohm resistor the voltage drop across the resistor will be:

- a) 100 mill volts
- b) 10 mill volts
- c) 1 volt
- d) 10 volts

FINAL TERM EXAMINATION

Spring 2003

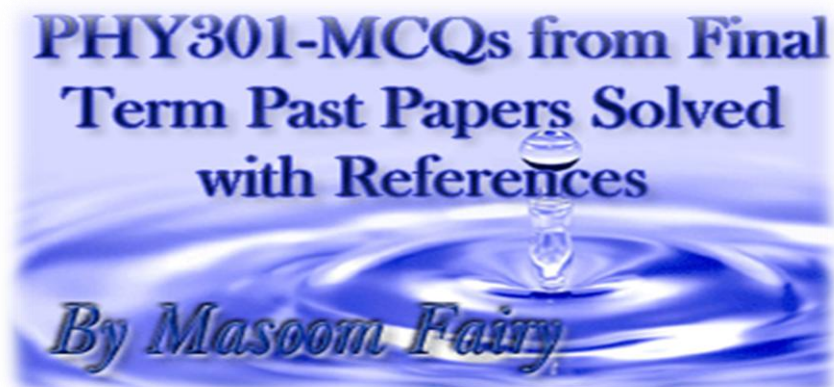
PHY301- Circuit Theory

International system of units is built upon how many basic units.

- ☐ 8
- ☐ 9
- ☐ 10
- ☒ 7 (PAGE 1)

Absolute zero corresponds to

- ☐ 0oC
- ☐ 32oC
- ☐ 273K



o 0 K (PAGE 1)

Two unlike charges will

o Attract (PAGE 9)

- o Repel
- o Both attract and repel
- o None of the above

With zero potential difference across a wire, the current will be

- o Maximum
- o Depends upon the wire

o Zero (PAGE 9)

- o None of the above

The opposition which limits the current is called

- o Conductance

o Resistance (PAGE 10)

- o Resistivity
- o Conductivity

The process whereby charged particles move under the influence of electric field is called

- o Current
- o Voltage

o Drift (PAGE 121)

- o None of the above

Phosphorus is a ----- element

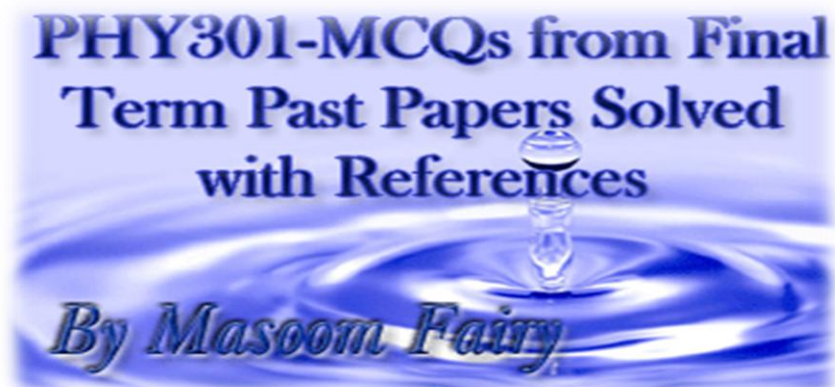
o Pentavalent (PAGE 122)

- o Trivalent
- o Hexavalent
- o None of the above

The current flows through the diode when it is reverse biased is called

- o Leakage current
- o Breakdown current
- o Saturation current

o None of the above (According to definitions of all above options)



The resistance offered by the inductor is called

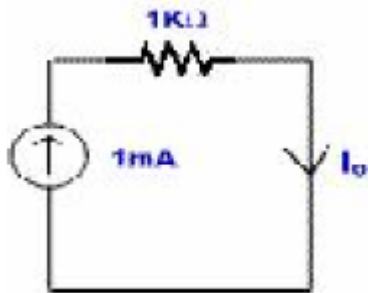
o Inductance (PAGE 19)

o Capacitance

o Resistivity

o None of the above

In the above circuit value of I_0 will be



o 2mA

o 1A (Current will be same in Series circuit)

o 2A

o 1mA

FINALTERM EXAMINATION

Spring 2006

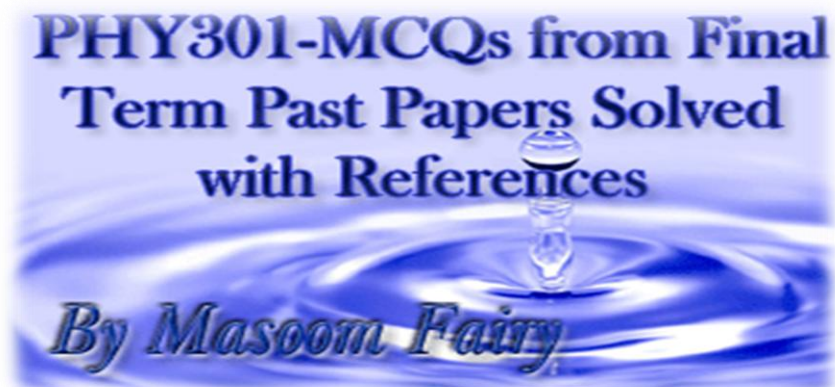
PHY301- Circuit Theory

Superposition theorem can be applied only to circuit having elements.

► Non-linear

► Passive

► Linear bilateral (PAGE 92)



- ▶ Resistive

The output of a half wave rectifier is suitable only for.....

- ▶ Running car diodes
- ▶ Charging batteries
- ▶ Diode rating
- ▶ Purity of power output

While calculating R_{th} current sources in the circuit are

- ▶ Replaced by opens (PAGE 100)

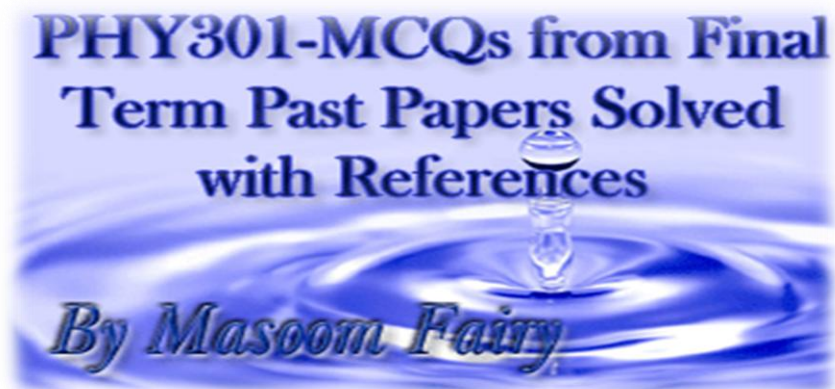
- ▶ Replaced indirect voltage
- ▶ Replaced by short
- ▶ Replaced DC voltage

In parallel circuits all components must

- ▶ have the same value
- ▶ have same potential difference (PAGE 34)
- ▶ carry the same current
- ▶ be switched ON & OFF simultaneously

The clipping level is primarily determined by

- ▶ shape of the input waveform
- ▶ value of the resistor



► battery voltage

► **knee voltage of the diode (PAGE 160)**

(2) Determine the applied voltage (Transition Voltage) that causes change in the diode bias.

The depletion region of a P-N junction is formed

► During the manufacturing

► **When forward bias is applied to it (PAGE 123)**

► Under reverse bias

► When its temperature is reduced

► None of the given

The dc load line of a transistor can be drawn if we know its cut off and points

► active

► **saturation**

► quiescent

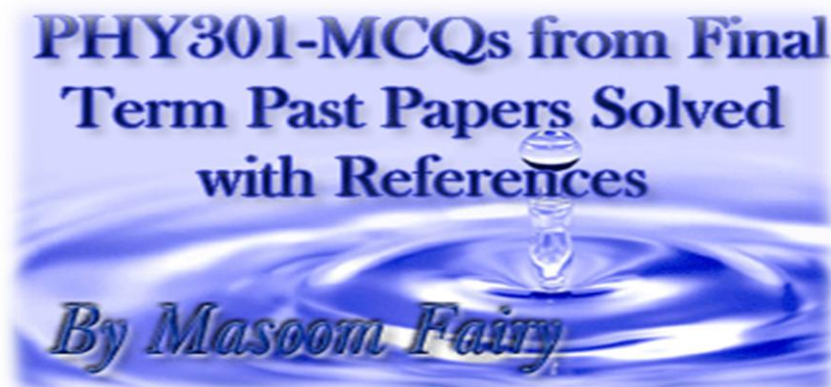
► None of them

Reference:

DC load line can be drawn onto these output characteristics curves to show all the possible operating points of the transistor from fully "ON" to fully "OFF".

http://www.electronics-tutorials.ws/amplifier/amp_2.html

The filter is simply a capacitor connected in ----- with the load resistance.



► series

► **parallel (PAGE 159)**

► we can use it in series as well as in parallel

► none of them

Zener diode is operated in the Region

► cut-off

► forward

► **breakdown**

► saturation

Zener diodes are a special type of semiconductor diode– devices that can be operated be operated continuously in that breakdown mode.

http://en.wikipedia.org/wiki/Zener_diode

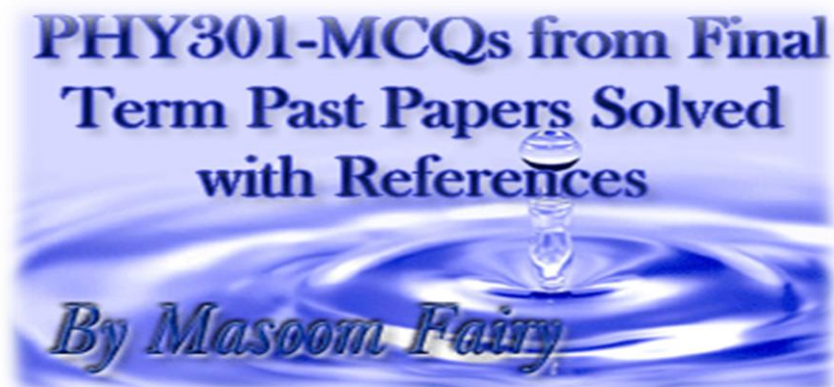
A tunnel diode is always biased

► By dc source

► In the reverse direction

► In the middle of its negative resistance region

► **None of them**



رہی تعلیم مشرق میں، نہ مغرب میں رہی غیرت
یہ جتنے پڑھتے جاتے ہیں، جہالت بڑھتی جاتی ہے!

PHY301-MCQs from Final
Term Past Papers Solved
with References

By Masoom Fairy

FINALTERM EXAMINATION
Fall 2009
PHY301- Circuit Theory (Session - 2)

Time: 120 min
Marks: 70

Question No: 1 (Marks: 1) - Please choose one

Charge of 2c and 5c will

- ▶ attract each other
- ▶ **repel each other**
- ▶ no effect
- ▶ cancel each other

Question No: 2 (Marks: 1) - Please choose one

Which battery applies a greater potential difference?

- ▶ **12v**
- ▶ 1.5v
- ▶ 10v
- ▶ 0.5v

Question No: 3 (Marks: 1) - Please choose one

When one resistance in a series string is open

- ▶ The current is maximum in the normal resistances.
- ▶ **The current is zero in all resistances**
- ▶ The voltage is zero across the open resistance
- ▶ The current increases in voltage source

Question No: 4 (Marks: 1) - Please choose one

Addition of impurities in semiconductor material to produce more current is called

- ▶ **dopping**
- ▶ bonding
- ▶ excitation
- ▶ intrinsic

PHY301 FINALTEM PAPER by:

Waseem.jeetumoni@gmail/yaho/skype/facebook.com

Question No: 5 (Marks: 1) - Please choose one

For a P-N junction under forward bias

- ▶ **more forward current flows**
- ▶ no forward current flows
- ▶ more reverse current flows
- ▶ infinite reverse current flows

Question No: 6 (Marks: 1) - Please choose one

A diode is a

- ▶ **one way conductor**
- ▶ two way conductor
- ▶ three way conductor
- ▶ insulator

Question No: 7 (Marks: 1) - Please choose one

In forward biased Characteristics of junction diode, current value is

- ▶ $i = I_s e^{V/V}$
- ▶ **$I_s = i e^{V/V T}$**
- ▶ $i = I_s e^{V T/V}$
- ▶ $i = I_s e^{V/V T}$

Question No: 8 (Marks: 1) - Please choose one

The primary and secondary winding of transformer are

- ▶ physically touched
- ▶ **physically isolated**
- ▶ touched with conductor
- ▶ largely separated

Question No: 9 (Marks: 1) - Please choose one

A certain transformer has 500 turns in the primary winding and 2500 turns in the secondary winding. The turn ratio is

- ▶ 0.5
- ▶ **25**
- ▶ 25v
- ▶ 2500

Question No: 10 (Marks: 1) - Please choose one

A transformer will be out of phase when

- ▶ output voltage is 0 degree out of phase with input voltage
- ▶ **output voltage is 180 degree out of phase with input voltage**
- ▶ output voltage is 360 degree out of phase with input voltage
- ▶ output voltage is same as input voltage

Question No: 11 (Marks: 1) - Please choose one

If N denotes the total number of nodes, then number of nodal equations in nodal analysis will be

- ▶ **Number of equations=N-1**
- ▶ Number of equations=N-1/2
- ▶ Number of equations=N
- ▶ Number of equations=N-2

Question No: 12 (Marks: 1) - Please choose one

Using superposition theorem, for a circuit containing independent sources, any remaining current source is

- ▶ replaced by capacitor
- ▶ replaced by short circuit
- ▶ **made zero by replacing them by open circuit**
- ▶ replaced by close circuit

Question No: 13 (Marks: 1) - Please choose one

The PIV of a half wave rectifier circuit with a shunt capacitor filter is

- ▶ $2V_m$
- ▶ V_m
- ▶ $V_m/2$
- ▶ $3V_m$

Question No: 14 (Marks: 1) - Please choose one

The primary function of a filter is to

- ▶ Minimize ac input variations
- ▶ Suppress odd harmonics in the rectifier output
- ▶ **Stabilize dc level of the output voltage**
- ▶ Remove ripples from the rectified output

Question No: 15 (Marks: 1) - Please choose one

The base region of a p-n-p transistor is

- ▶ Very thin and heavily doped with holes
- ▶ **Very thin and heavily doped with electrons**
- ▶ Very thin and lightly doped with holes
- ▶ Very thin and lightly doped with electrons

Question No: 16 (Marks: 1) - Please choose one

Voltage multipliers are usually used in low current high voltage applications e.g

- ▶ Stabilizer of refrigerator
- ▶ Microcontroller
- ▶ **Cathode ray tube in television**
- ▶ Remote control

Question No: 17 (Marks: 1) - Please choose one

The current flow across the base-emitter junction of a p-n-p transistor consists of

- ▶ Mainly electrons
- ▶ Equal numbers of holes and electrons
- ▶ **Mainly holes**
- ▶ The leakage current

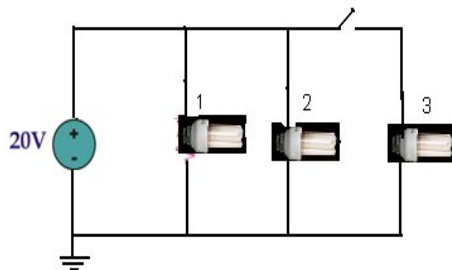
Question No: 18 (Marks: 1) - Please choose one

In an P- type semiconductor material , the majority carriers are

- ▶ **holes**
- ▶ electrons
- ▶ ions
- ▶ protons

Question No: 19 (Marks: 1) - Please choose one

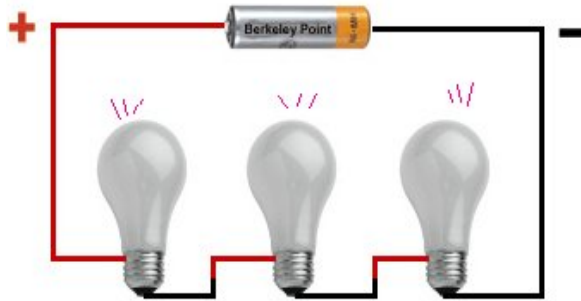
Which light bulb will not glow in given circuit



- ▶ bulb 1 and 2
- ▶ all bulbs will not glow
- ▶ all bulbs will glow
- ▶ **bulb 3**

Question No: 20 (Marks: 1) - Please choose one

Three bulbs are connected in series of a battery, what would happen if any one bulb is opened.



- ▶ half of current will flow
- ▶ same current will flow
- ▶ **no current will flow**
- ▶ $\frac{2}{3}$ current will flow

Question No: 21 (Marks: 3)

A small light bulb with a resistance of 25Ω is connected across the same 220 volt power line. How much is the current 'I'

Solution:

$$R=25\ \Omega$$

$$V=220\ \text{V}$$

From ohm law

$$V=IR, \quad I=V/R$$

So,

$$I=220/25=8.8\ \text{Ampere}$$

Question No: 22 (Marks: 3)

Describe Source Transformation method for simplifying circuit..

Source Transformation:

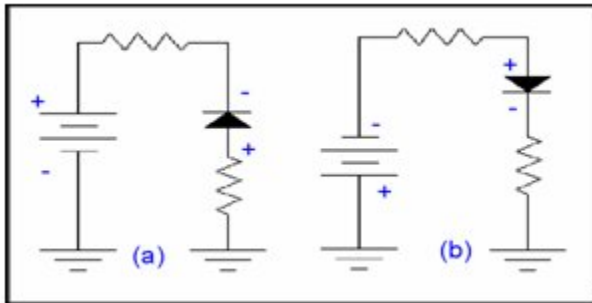
If we have any source embedded within a network, say this source is a current source having a value I & there exists a resistance having a value R , in parallel to it. We can replace it with a voltage source of value $V=IR$ in series with same resistance R .

The reverse is also true that is a voltage source V , in series with a resistance R can be replaced by a current source having a value $I= V/R$ in parallel to the resistance R .

Parameters within circuit, for example an output voltage remain unchanged under these transformations.

Question No: 23 (Marks: 3)

Given below are two figures (a) and (b) having Diode, which diode is forward biased or reversed biased, tell reason.



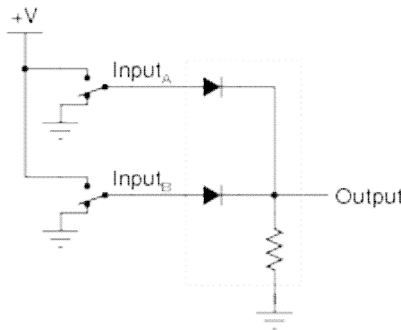
Both A and B are reversed biased

In **fig. a** the negative of the battery is connected to the anode through a resistor and positive terminal of the battery is connected the cathode therefore no current will flow and we can say that diode is not existing and it will act as an open circuit.

In **fig. b** the negative terminal of the battery is connected to the anode through a resistor and positive terminal of the battery is connected the cathode

Question No: 24 (Marks: 3)

Crude logic gates circuits may be constructed out of nothing but diodes and resistors. Take for example this logic gate circuit:



Identify what type of logic function is represented by this gate circuit

This is an OR gate circuit.

Question No: 25 (Marks: 3)

Differentiate between Half wave & Full Wave Rectifier.

Half Wave Rectifier:

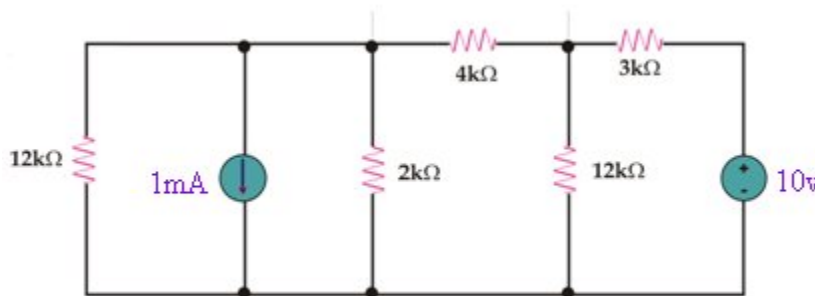
A half wave rectifier is a special case of a clipper. In half wave rectification, either the positive or negative half of the AC wave is passed easily, while the other half is blocked, depending on the polarity of the rectifier. Because only one half of the input waveform reaches the output, it is very inefficient if used for power transfer. Half-wave rectification can be achieved with a single diode in a one phase supply.

Full Wave Rectifier:

A full-wave rectifier converts the whole of the input waveform to one of constant polarity (positive or negative) at its output. Full-wave rectification converts both polarities of the input waveform to DC (direct current), and is more efficient. However, in a circuit with a non-center tapped transformer, four diodes are required instead of the one needed for half-wave rectification.

Question No: 26 (Marks: 5)

Keeping in mind the **Source transformation method**, how we will convert **10v** voltage source into current source and **1mA** into voltage source in the following circuit, Draw diagrams of converted circuit.



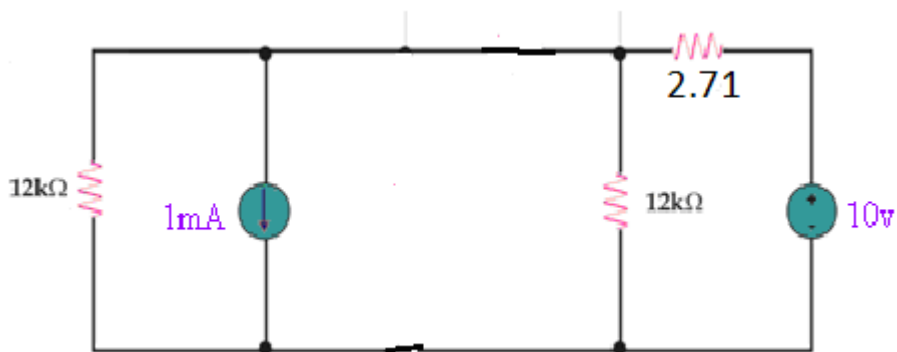
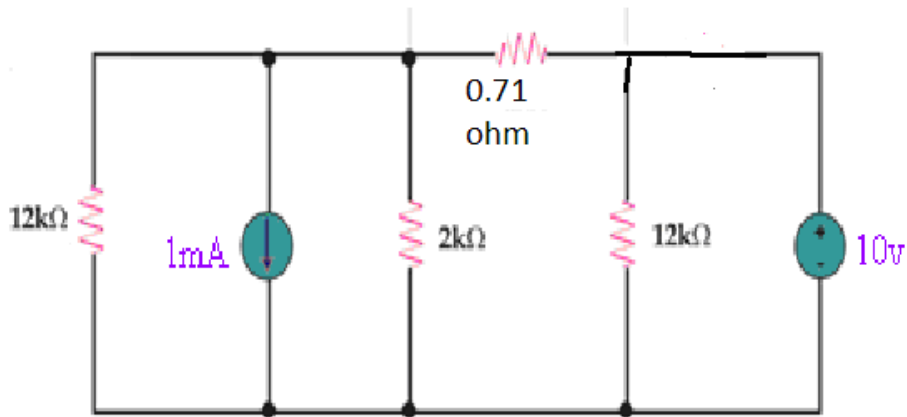
4 is in parallel with 3 so it becomes

$$4 \cdot 3 / 4 + 3 = 1.71 \text{ ohm}$$

0.71 again in series with 2 to give 2.71 ohm

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1mA multiply with 12 k by source transformation to give 12 volts

<http://vuattach.ning.com>

2.71 is in series with 12 to give 12.71 .

10 V is in series with 2.71 so, by source transformation

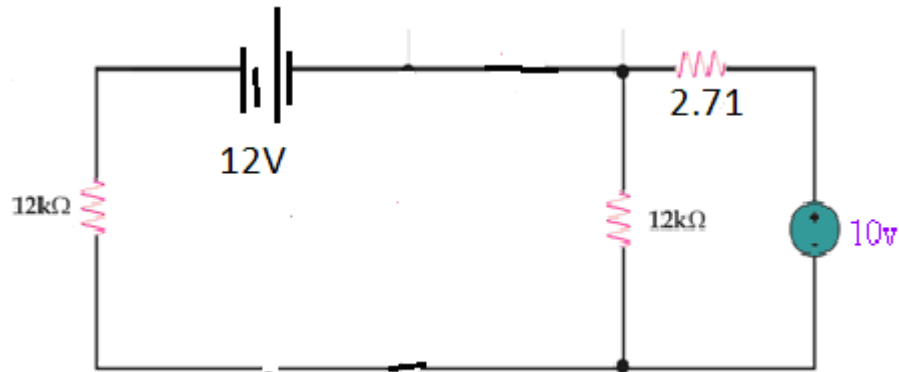
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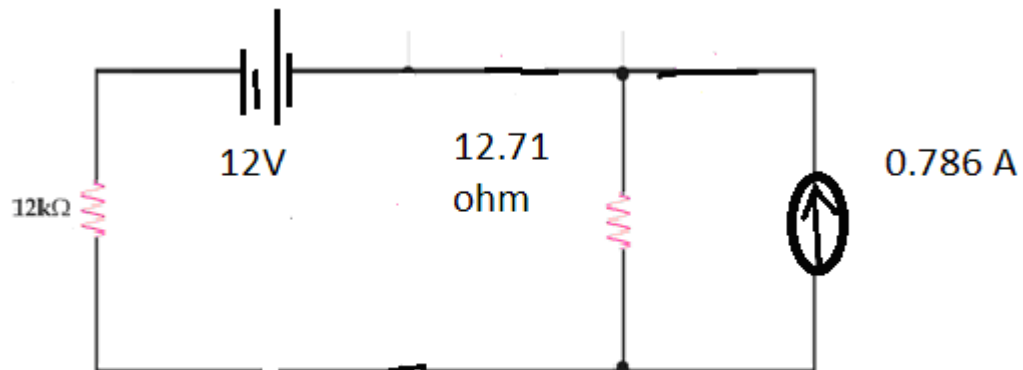
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So ,



$$I = 10/12.71 = 0.786A$$

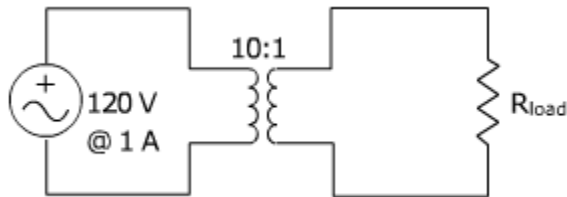
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Question No: 27 (Marks: 5)

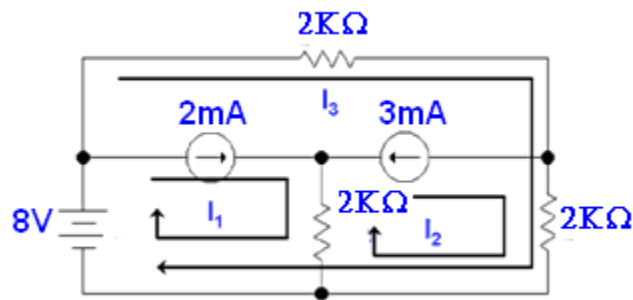
Did the power remain same or not, in either case provide a proof?
<http://vuattach.ning.com>



The power will not remain same. if primary side is equal to 120V then the secondary side will be
 $120/10$
 $=12V$

Question No: 28 (Marks: 5)

Calculate I_1 , I_2 and I_3 values.



Solution:

$$I_1 = 2\text{mA} \dots \text{Ans}$$

$$I_2 = -3\text{mA} \dots \text{Ans}$$

KVL for loop 3

$$2kI_3 + 2k(I_3 + I_2) = 8$$

$$2kI_3 + 2kI_3 + 2kI_2 = 8$$

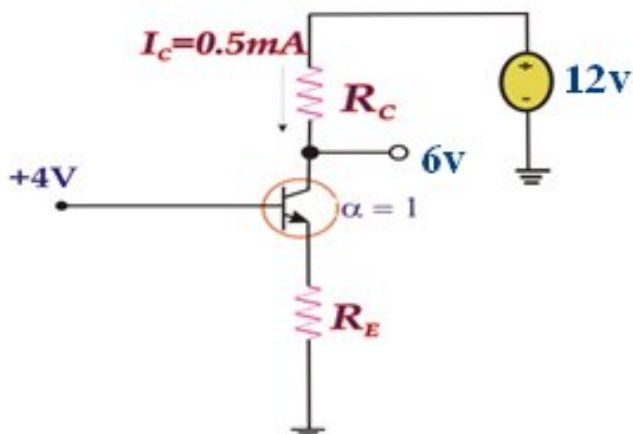
$$4kI_3 - 8 = 8$$

$$4kI_3 = 16$$

$$I_3 = 4\text{mA} \dots \text{Ans}$$

Question No: 29 (Marks: 5)

For the circuit shown, find I_E & R_C



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Question No: 30 (Marks: 5)

A particular diode, for which $n=1$, is found to conduct 3mA with a junction voltage of 0.7V to 0.8V. Find diode current..

Solution:

$$I_d = I_s e^{V_d / nV_t}$$

$$\text{If } I_d = 3 \times 10^{-3} \text{ A}$$

$$\text{when } V_d = 0.7$$

then

$$3 \times 10^{-3} = I_s e^{0.7 / 25.2 \times 10^{-3}}$$

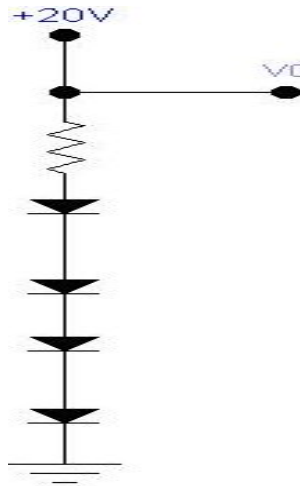
$$= I_s = 2.5905 \times 10^{-15}$$

$$\text{when } V_d = 0.7 \quad I_d = 4.4619 \text{ mA}$$

$$\text{when } V_d = 0.8 \quad I_d = 0.15868 \text{ mA}$$

Question No: 31 (Marks: 10)

Design the circuit in the fig. so that $V_O = 3V$ when $I_L = 0$, and V_O changes by 40mV per 1mA of load current. Find the value of R . (assume four diodes are identical) relative to a diode with 0.7v drop at 1 mA current. Assume $n = 1$



<http://vuattach.ning.com>

SOLUTION: on next:

$V_O = 3V$, when $I_L = 0$, therefore each diode should exhibit a drop of 0.75V. If $I_L = 1mA$, then V_O changes by 40mV and a change due to each diode is 10mV.

Hence

$$r_d = 10mV/1mA = 10 \text{ Ohms}$$

but

$$r_d = nV_T/I_D$$

$$10 = 1 \times 25m/I_D$$

$$I_D = 2.5mA$$

Hence

$$15 - 3 - I_D R_D = 0$$

$$R = (15 - 3)/I_D = (15 - 3)/2.5 = 4.8k \text{ Ohms}$$